

$$= 2.1 \text{ J}$$

10	C If the roller coaster have lost contact, the normal contact force of track on roller coaster must be zero.	D
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	Centripetal force will be provided by component of weight pointing towards O, if it just remains in contact. At position P, Component of weight towards O = $mg \cos 30^\circ$ To maintain contact, $mg \cos 30^\circ = \frac{mv^2}{r}$ $v = \sqrt{rg \cos 30^\circ}$ $= \sqrt{1.0(9.81) \cos 30^\circ}$ $= 2.91 \text{ m s}^{-1}$ So, option C will be the best option.	
11	C	A